

Group_03 — Progress Review I: Team Guide

Dataset: Seoul Bike Sharing Demand

Read this before you start your part. Every member follows the same structure.

1. What We're Submitting

One zip file to Courseweb (Group_03.zip) containing: each member's individual notebook, a combined group_pipeline.ipynb, the dataset, our EDA charts, and a README. Every member is individually marked (20 marks) on their own notebook + viva presentation. There are 5 shared marks for how well the group_pipeline.ipynb integrates everyone's work together.

2. Dataset Basics (same for everyone)

File: **data/raw/SeoulBikeData.csv** — 8,760 rows, 14 columns, one row per hour (Dec 2017 to Nov 2018).
Target column: **Rented Bike Count**.

Important: the file is NOT UTF-8 encoded. Load it like this or you'll get an error:

```
df = pd.read_csv('data/raw/SeoulBikeData.csv', encoding='cp1252')
```

There are no missing values and no duplicate rows in the raw data (already checked). Three columns are text/categorical: Seasons, Holiday, Functioning Day — everything else is numeric.

3. Pick Your Technique (one per member, no repeats)

Technique	What it does	Status
Encoding Categorical Variables	Convert Seasons/Holiday/Functioning Day text into numbers	DONE (see notebooks/)
Handling Missing Data	Check & handle missing values (dataset has none, so this member should demonstrate the checking process and justify why no imputation was needed)	Open
Outlier Removal	Detect & handle extreme values (e.g. Rented Bike Count, Wind speed, Rainfall, Snowfall look skewed)	Open
Normalization / Scaling	Scale numeric columns (Temperature, Humidity, Wind speed, Visibility, Dew point, etc.) to a common range	Open
Feature Engineering	Create/select features — e.g. extract day-of-week from Date, cyclical encoding of Hour, correlation check between Temperature and Dew point	Open

If your group has 6 members, split Feature Engineering into two parts (e.g. one does feature selection/correlation analysis, the other does dimensionality reduction), or add a second angle to any technique above (e.g. two different outlier-detection methods).

4. Folder Structure & File Naming

```
Group_03/  
|-- README.md  
|-- data/raw/SeoulBikeData.csv  
|-- notebooks/  
|   |-- IT_Number_Encoding_Categorical_Variables.ipynb  
|   |-- IT_Number_Handling_Missing_Data.ipynb
```

```
|      |-- IT_Number_Outlier_Removal.ipynb
|      |-- IT_Number_Normalization_Scaling.ipynb
|      `-- IT_Number_Feature_Engineering.ipynb
|-- group_pipeline.ipynb
`-- results/
    |-- eda_visualizations/    (save your chart as .png here)
    |-- outputs/               (save your processed CSV here)
    `-- logs/
```

Replace **IT_Number** with your real IT number, e.g. IT21123456_Outlier_Removal.ipynb. Don't rename the folders themselves.

5. What Must Be Inside Your Individual Notebook

Every member's notebook must have these 4 things, in order — this is exactly what's marked in the viva:

- **1. Explain the technique** — a markdown cell describing what the technique does, in plain words.
- **2. Justify why it's needed for THIS dataset** — don't give a generic textbook reason. Point at something specific in the Seoul Bike data (e.g. “Wind speed has extreme values above 7 m/s”, not just “outliers can affect models”).
- **3. Show implementation** — actual working code cells with real output shown underneath (not just code with no output).
- **4. One EDA visualization + interpretation** — a chart (histogram, boxplot, correlation matrix, bar chart, etc.) plus 2-3 sentences explaining what it shows and why it matters.

Follow notebooks/IT_Number_Encoding_Categorical_Variables.ipynb as a template for structure and tone — copy its markdown cell style, its section headers, and the way it explains reasoning before showing code.

Load the dataset the same way at the top of your notebook so everyone's notebook can run independently:

```
import pandas as pd
df = pd.read_csv('data/raw/SeoulBikeData.csv', encoding='cp1252')
```

6. How the Combined group_pipeline.ipynb Works

This is a SEPARATE notebook (already started) that runs everyone's technique together, in the correct order, on one shared dataframe. It's what gets the 5 shared group marks — the marker is checking that our steps connect logically and don't undo each other.

Fixed order (don't change this):

1. Handling Missing Data
2. Encoding Categorical Variables
3. Outlier Removal
4. Normalization / Scaling
5. Feature Engineering

Why this order matters: Encoding must happen before Scaling, because the new 0/1 and one-hot columns should NOT be scaled — scaling them would distort their meaning. Outlier removal should generally happen on numeric columns after encoding, and before scaling changes their range.

How to add your part:

- Open group_pipeline.ipynb — you'll see a markdown heading for each step (e.g. “Step 3 — Outlier Removal”) with a TODO code cell under it.
- Copy the working code from YOUR OWN notebook (not the explanation text, just the code) into your step's TODO cell.
- Do not touch other members' sections.
- Run the whole notebook top to bottom once everyone's code is in, to confirm nothing breaks in sequence.
- The last cell saves the final processed dataset to results/outputs/ — this only needs to run once, at the end.

Whoever adds a scaling or outlier-removal step: exclude the Season_*, Holiday, and Functioning Day columns from your operation — they're already 0/1 encoded by the time your step runs.

7. Before We Zip & Submit

- [] Every member's notebook is renamed with their real IT number.
- [] Every member's notebook runs top-to-bottom without errors.
- [] README.md member table is filled in with real names/IT numbers/techniques.
- [] group_pipeline.ipynb runs top-to-bottom with everyone's code integrated, in the correct order.
- [] Charts are saved in results/eda_visualizations/, final dataset in results/outputs/.
- [] Zip the whole Group_03 folder (not just the files inside it) and upload to Courseweb.

Questions about your specific technique? Ask in the group chat before the night before the viva — not during it.